Jackfruit Phase-1

Software Requirements Specification

for

Railway Reservation System

Version 1.0

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Date: 05-09-2026

Revision history

Name	Date	Reason for Changes	Version
Narendra Kathare	03-09-2026	• Initial draft: project scope, purpose, overall description • Final review — Non-functional requirements section	1.1
Yellapantula Venkata Pranav	04-09-2026	• External interface requirements + system features • Final review — Formatting check	1.2
Yashash C	05-09-2026	• Analysis models: layered architecture, use case diagrams, sequence diagrams • Final review — Consistency check (cross-referencing all sections)	1.3

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Railway Reservation System. It defines functional and non-functional requirements, interfaces, and verification criteria intended for instructors and students to use as a reference.

1.2 Scope

Covers passenger interactions for searching, booking, paying for, and tracking train tickets, plus administrative functions for managing trains, schedules, fares, cancellations, and reports. Excludes integration with real railway core systems (such as actual IRCTC/Indian Railways infrastructure) and real payment gateway processing beyond a simulated/mock module.

1.3 Audience

Developers, QA Engineers, System Integrators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: RRS (Railway Reservation System), PNR (Passenger Name Record), RAC (Reservation Against Cancellation), UI (User Interface), API (Application Programming Interface), DB (Database), TLS (Transport Layer Security), CLI (Command-Line Interface).

2. Overall description

2.1 Product perspective

The Railway Reservation System is a self-contained, web-based application used by passengers and administrators. It includes a booking front end, a seat-allocation and fare-processing service, and a MySQL-backed data store for trains, schedules, bookings, and payments. It is a new academic project and is not connected to any real railway operator's infrastructure.

2.2 Major product functions (detailed)

- Authenticate user (registration and login)
- Search trains by source, destination, and date
- Check seat/berth availability by class
- Book a ticket (select class, berth preference, passenger details)
- Process payment (simulated) and confirm booking
- Generate PNR and support PNR-based status tracking
- Cancel a booking and process a refund
- Smart seat allocation: automatic waitlist / RAC promotion
- Administrative maintenance: manage trains, schedules, fares, and quotas
- Reports and analytics for bookings, occupancy, and revenue

2.3 User roles and characteristics (expanded)

- Passenger: basic digital literacy, expects a short, clear booking flow and visibility into booking status.
- Administrator: manages trains/schedules and monitors bookings; needs a dashboard with clear, actionable information.
- Evaluator/Instructor: reviews design quality and code structure; expects clean separation between booking, allocation, storage, and presentation layers.

2.4 Operating environment

Runtime: Python 3.x with Flask. Database: MySQL. Client: any modern web browser or a terminal for the CLI build. No special hardware or embedded OS is required.

2.5 Constraints

- Booking, allocation, storage, and presentation must remain separate, swappable layers behind defined interfaces
- Payment is handled by a simulated/mock module only — no real payment gateway is integrated for this course project
- Only synthetic/sample train, schedule, and passenger data is used — no real personal data
- Must run comfortably on a standard student laptop/development server

3. External interface requirements

3.1 User interfaces

Primary UI: a simple web form (or CLI) with a clear menu structure — search, book, pay, track PNR, cancel. Accessibility: legible fonts, clear labels, and high-contrast, readable status indicators (Confirmed / RAC / Waitlisted / Cancelled).

3.2 Hardware interfaces

None required for the core system. All functions are software-based; no card readers, printers, or dedicated kiosks are needed for this course project.

3.3 Software interfaces

- Application Backend (Flask, Python) — serves booking, allocation, and reporting logic over HTTP(S)/JSON
- MySQL Database — stores users, trains, schedules, bookings, and payment records via a repository-style interface
- Mock Payment Module — simulates a payment gateway call and returns a Success/Failure status
- Monitoring/Reporting module — for admin dashboard statistics and exports

3.4 Communications

- HTTPS / TLS 1.2+ enforced for all client–server communication in deployment
- No calls to external, real-world railway or payment services; all external interactions are simulated for the course project

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow RRS-F-###.

4.1 Authentication

Description: Authenticate the passenger (registration and login) and restrict access to protected functions accordingly.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-001	The system shall allow a new user to register and validate the registration information before creating an account.	Functional	High	Security	AC-RRS-F-001: Valid registration data creates an account; duplicate email is rejected. Test: TC-Auth-01	Requires DB
RRS-F-002	The system shall authenticate a registered user using valid login credentials.	Functional	High	Security	AC-RRS-F-002: Correct credentials grant access to the dashboard. Test: TC-Auth-02	—
RRS-F-003	The system shall deny access after repeated failed login attempts and display an appropriate message.	Functional	High	Security	AC-RRS-F-003: After 5 failed attempts, further attempts are blocked temporarily. Test: TC-Auth-03	Rate-limit config

4.2 Train Search

Description: Let a passenger search for trains by source, destination, and date of travel.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-010	The system shall let a user search for trains by source station, destination station, and date of travel.	Functional	High	Business	AC-RRS-F-010: Matching trains are returned with class and timing. Test: TC-Search-01	Requires seed schedule data

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-011	The system shall display a clear message when no trains match the search criteria.	Functional	Medium	Business	AC-RRS-F-011: 'No trains found' message shown for an unmatched route/date. Test: TC-Search-02	—

4.3 Seat Availability and Booking

Description: Show live seat/berth availability and let the passenger create a booking.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-020	The system shall display current seat/berth availability for a selected train and class before booking.	Functional	High	Operations	AC-RRS-F-020: Availability count matches the allocation service state. Test: TC-Book-01	Allocation service
RRS-F-021	The system shall let the user enter passenger details and create a provisional booking pending payment.	Functional	High	Business	AC-RRS-F-021: Provisional booking is created and held for a limited time. Test: TC-Book-02	Session/timeout config
RRS-F-022	The system shall validate all required booking fields and reject incomplete submissions with a clear error.	Functional	High	QA	AC-RRS-F-022: Missing/invalid fields are flagged before submission proceeds. Test: TC-Book-03	—

4.4 Payment Processing

Description: Process payment for a provisional booking through the mock payment module and confirm the booking on success.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-030	The system shall process ticket payment through the mock payment module and confirm the booking only on success.	Functional	High	Business	AC-RRS-F-030: Booking status becomes Confirmed only after a Success payment response. Test: TC-Pay-01	Mock payment module
RRS-F-031	The system shall release the provisional booking and any held seat if payment fails or times out.	Functional	High	Operations	AC-RRS-F-031: Failed/timed-out payment releases the seat back to availability. Test: TC-Pay-02	Allocation service

4.5 PNR Tracking, Cancellation and Refund

Description: Let a passenger track a booking by PNR and cancel it, with an associated refund calculation.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-040	The system shall generate a unique PNR for every confirmed booking and allow lookup by PNR.	Functional	High	Business	AC-RRS-F-040: A confirmed booking always returns a valid PNR; lookup returns correct status. Test: TC-PNR-01	—
RRS-F-041	The system shall allow a user to cancel a booked ticket after confirming the action.	Functional	High	Business	AC-RRS-F-041: Confirmed cancellation updates status to Cancelled and releases the seat. Test: TC-Cancel-01	Allocation service
RRS-F-042	The system shall calculate and record the applicable refund	Functional	Medium	Business	AC-RRS-F-042: Refund amount matches the configured	Policy config

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	amount according to the cancellation policy.				cancellation-policy table. Test: TC-Cancel-02	

4.6 Smart Seat Allocation (Waitlist / RAC)

Description: Automatically manage waitlist and RAC promotion when confirmed seats become available.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-050	The system shall automatically assign a booking status (Confirmed / RAC / Waitlisted) based on available capacity at the time of booking.	Functional	Medium	Operations	AC-RRS-F-050: Status assigned matches current capacity for the train/class/date. Test: TC-Alloc-01	Allocation service
RRS-F-051	The system shall automatically promote the next eligible waitlisted/RAC passenger when a confirmed seat becomes available, in booking order.	Functional	Medium	Operations	AC-RRS-F-051: A cancellation triggers promotion of the earliest eligible waitlisted booking. Test: TC-Alloc-02	Requires event trigger on cancellation

4.7 Admin: Train/Schedule Management and Reports

Description: Provide administrators with train/schedule management and reporting functions.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
RRS-F-060	The system shall let an authorized administrator add, update, or	Functional	High	Operations	AC-RRS-F-060: Schedule changes are reflected	Admin auth required

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	remove trains and schedules.				immediately in search results. Test: TC-Admin-01	
RRS-F-061	The system shall let an authorized administrator generate booking, occupancy, and revenue reports.	Functional	Medium	Business	AC-RRS-F-061: Report figures match underlying booking data for the selected period. Test: TC-Admin-02	Reporting module

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow RRS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
RRS-NF-001	Overall booking transaction response (from search to confirmation) shall be ≤ 5 seconds for 90% of transactions under normal load.	Performance	High	90th percentile ≤ 5s in test environment. Test: TC-Perf-01
RRS-NF-002	System shall provide 99% availability during scheduled demo/evaluation periods, excluding planned maintenance.	Reliability	High	Uptime logs show ≥99% during evaluation window. Test: Ops logs
RRS-NF-003	All sensitive user and payment data must be protected; passwords and card-like data must never be stored in plaintext.	Security/Compliance	High	Security review checklist pass. Test: TC-Sec-01
RRS-NF-004	System shall retain booking and audit logs with timestamped events for the duration of the academic evaluation period.	Audit/Data Retention	Medium	Log retrieval verified for sample bookings. Test: TC-Ops-01
RRS-NF-005	System shall provide a usable interface with clear labels and sensible defaults, requiring no manual configuration to get started.	Usability	Medium	Usability walkthrough completed without external help. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

- Protect passenger personal and payment-related data from unauthorized access or disclosure.
- Ensure booking and payment transactions are authenticated and cannot be silently tampered with.
- Prevent unauthorized administrative actions on train, schedule, and fare data.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
PRJ-SR-001	TLS 1.2+ mandatory for all network connections in deployment.	Security	High	AC-SR-001: Connections without TLS are rejected. Test: TC-Sec-02
PRJ-SR-002	Passwords shall be hashed (never stored in plaintext).	Security	High	AC-SR-002: Database inspection shows only hashed values. Test: TC-Sec-03
PRJ-SR-003	Role-based access control shall be enforced at the server, not just hidden in the UI.	Security	High	AC-SR-003: Direct API calls to admin-only endpoints are rejected for a user role. Test: TC-Sec-04

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
PRJ-SR-004	User sessions shall expire after a defined period of inactivity.	Security	Medium	AC-SR-004: Session token is invalidated after the configured idle timeout. Test: TC-Sec-05
PRJ-SR-005	The mock payment module shall not persist full card-like details; only a transaction reference and status shall be stored.	Security	High	AC-SR-005: Database contains no raw card-like fields. Test: TC-Sec-06
PRJ-SR-006	All administrative actions (schedule changes, refunds) shall be logged with user identity and timestamp.	Security	Medium	AC-SR-006: Audit log entry exists for every admin action performed in test. Test: TC-Sec-07

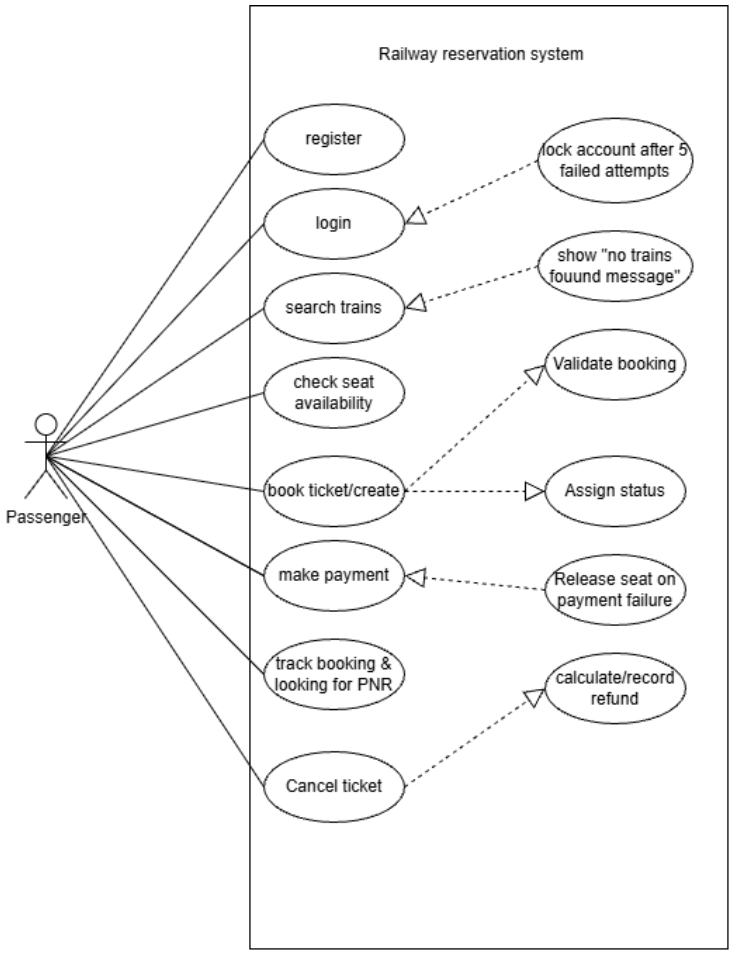
6. Quality attributes & Acceptance tests

Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all high-priority test cases passed.

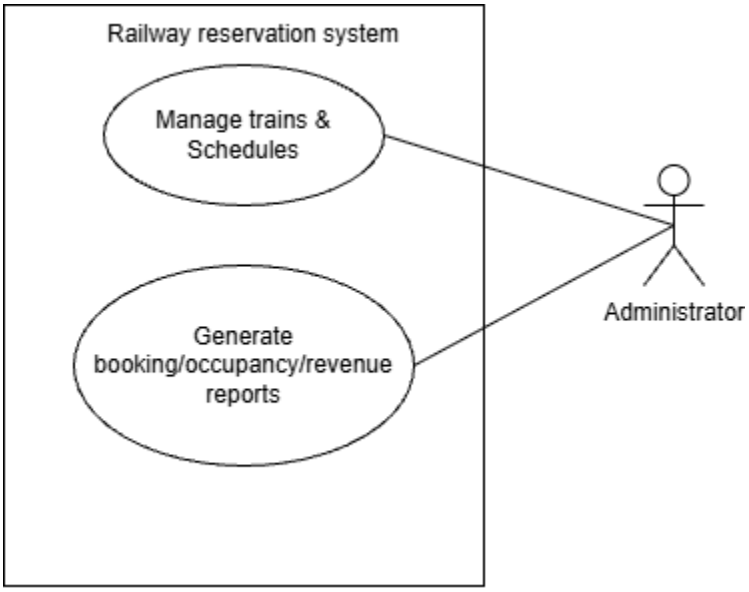
Acceptance test suites: Authentication, Train Search, Booking, Payment, Cancellation/Refund, Smart Allocation, Performance, Security, and Admin Reporting tests.

7. UML Use-Case Diagram

7.1 Passenger use case diagram



7.2 Administrator use case diagram



8. Requirements Traceability Matrix (RTM)

Status (N/P/A) = Not started / Partially implemented / Achieved (passed). Status is 'N' throughout since this is the design-stage SRS.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
RRS-F-001	Register / authenticate user	1.4 / DS-Auth-01	AuthModule	TC-Auth-01, TC-Auth-02	N	
RRS-F-010	Search trains by route/date	4.2 / DS-Search-01	SearchModule	TC-Search-01, TC-Search-02	N	
RRS-F-021	Create provisional booking	4.3 / DS-Booking-01	BookingModule	TC-Book-02, TC-Book-03	N	
RRS-F-030	Confirm booking after payment	4.4 / DS-Payment-01	PaymentModule	TC-Pay-01, TC-Pay-02	N	
RRS-F-040	Generate PNR / status lookup	4.5 / DS-PNR-01	BookingModule	TC-PNR-01	N	
RRS-F-041	Cancel booking, release seat	4.5 / DS-Cancel-01	BookingModule	TC-Cancel-01, TC-Cancel-02	N	
RRS-F-050	Auto seat status assignment	4.6 / DS-Alloc-01	AllocationModule	TC-Alloc-01	N	
RRS-F-051	Waitlist/RAC promotion	4.6 / DS-Alloc-02	AllocationModule	TC-Alloc-02	N	
RRS-F-060	Admin train/schedule management	4.7 / DS-Admin-01	AdminModule	TC-Admin-01	N	
RRS-NF-001	Response time target	5 / DS-Perf-01	WebUI / BackendAPI	TC-Perf-01	N	
PRJ-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	N	
PRJ-SR-003	Server-side RBAC	5.1.2 / DS-Sec-02	AdminModule	TC-Sec-04	N	